

HWA CHONG INSTITUTION



C2 PRELIMINARY EXAMINATION

9746 H2 CHEMISTRY

Paper 3 Free Response

18 September 2009

2 hours

Do not open this booklet until you are told to do so.

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Answer any **four** questions.

Begin each question on a **new piece of paper**.

A *Data Booklet* is provided.

You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

Write down the **question number** that you have answered **on the cover page** provided.

Answer any **four** questions.

- 1 (a) Liquid phosphorus trichloride is prepared by the reaction of chlorine with white phosphorus. Some related thermochemical data are shown below.

Enthalpy change of formation of $\text{PCl}_3(l)$	-320 kJmol^{-1}
Enthalpy change of vaporization of $\text{PCl}_3(l)$	$+32 \text{ kJmol}^{-1}$
Enthalpy change of atomisation of phosphorus	$+314 \text{ kJmol}^{-1}$

- (i) Using the data given, and relevant data from the *Data Booklet*, construct an energy level diagram to determine the P–Cl bond energy in PCl_3 .
- (ii) If an electric discharge is applied to a mixture of PCl_3 vapour and hydrogen gas, a rare chloride of phosphorus, diphosphorus tetrachloride (P_2Cl_4) is formed as one of the products.

Draw a dot-and-cross diagram of a molecule of P_2Cl_4 and state the shape with respect to each phosphorus atom in the molecule, indicating the bond angle.

[5]

- (b) The other product formed in the reaction in **a(ii)** is HCl . The HCl formed can react with but-1-ene to form 2-chlorobutane.

- (i) Describe the mechanism of this reaction.
- (ii) Write the rate equation of this reaction.
- (iii) Predict how the rate of reaction will be affected if HCl is replaced by HI , using relevant data from the *Data Booklet* to explain your answer.

[6]

- (c) With reference to the mechanism in **b(i)**, discuss the effect (if any) of the product obtained on plane-polarized light.

[4]

- (d) Although chlorine and iodine are elements in Group VII, they can differ in their physical properties and chemical reactivities.

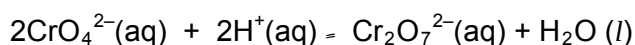
- (i) Compare the volatility of chlorine and iodine.
- (ii) Using relevant data from the *Data Booklet*, describe and explain with the aid of equations, how chlorine and iodine react differently with aqueous sodium thiosulfate.

[5]

[Total: 20]

2 This question is about the chemistry of chromate(VI) and dichromate(VI) compounds.

- (a) 8.00 g of potassium chromate is dissolved in acid to make a 100 cm³ solution. A *dynamic equilibrium* occurs, according to the equation below:



$$K_{\text{C}} = 7.55 \times 10^{12} \text{ mol}^{-3} \text{ dm}^9$$

- (i) What do you understand by the term *dynamic equilibrium*?
- (ii) Write the K_{C} expression for the above equilibrium and calculate the pH at which only **one-fifth** of the original amount of chromate ions remain.
- (iii) Predict and explain the pH change when aqueous BaCl₂ is added to the above mixture to precipitate BaCrO₄.

[6]

- (b) Lead(II) chromate was used for a long time as a yellow pigment in oil paintings.

- (i) Using the following standard reduction potentials:
- | | |
|---|-----------------------------|
| $\text{CrO}_4^{2-} + 8\text{H}^+ + 3\text{e}^- \rightleftharpoons \text{Cr}^{3+} + 4\text{H}_2\text{O}$ | $E^{\circ} = +1.33\text{V}$ |
| $\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{SO}_2 + 2\text{H}_2\text{O}$ | $E^{\circ} = +0.17\text{V}$ |

explain why the yellow color slowly changes when the paint is exposed to an atmosphere containing sulfur dioxide.

- (ii) Suggest why this colour change takes a long time, even though in the laboratory the reaction occurs immediately.
- (iii) State the colour change that takes place.

[4]

- (c) Acidified potassium dichromate(VI) can be used as an oxidizing agent in redox titrations and in organic reactions.

- (i) In a redox titration to determine the concentration of a sample of acidified potassium dichromate(VI) solution, 100 cm³ of potassium iodide solution was added to 100 cm³ of the sample to react with all the dichromate(VI) present. 25.0 cm³ of the resulting mixture required 21.50 cm³ of a 0.20 mol dm⁻³ of sodium thiosulfate solution for complete reaction. Calculate the concentration of the potassium dichromate(VI) in the sample.
- (ii) Reaction of **A**, C₄H₈O, with acidified potassium dichromate(VI) first yields **B**, and then **C**. **B** can undergo reaction to form a 2,4-dinitrophenylhydrazone. **B** and **C** give the same organic product CH₃CO₂H when treated with hot concentrated acidified potassium manganate(VII). However **B** and **C** give different products, **D** and **E** respectively, when reacted with hydrogen in the presence of platinum.

Deduce the structures of compounds **A** to **E**, as well as the structure of the 2,4-dinitrophenylhydrazone, explaining the chemistry of the reactions described.

[10]

Total: [20]

- 3 (a) Sketch a graph of the melting points of the elements in Period 3 from sodium to

argon, and explain the variations shown in terms of the structure and bonding present in the elements.

[3]

- (b) Beryllium is an atypical element in Group II, and it shows a diagonal relationship with aluminium, which is in Group III.

Justify this statement by comparing the reactions of beryllium hydroxide and magnesium hydroxide with hydrochloric acid and also with sodium hydroxide. Write equations, including state symbols, to illustrate your answer.

[3]

- (c) The Group II compounds, magnesium sulfate and barium sulfate, dissolve in water to different extent.

- (i) Using the data below, calculate the standard Gibbs' free energy change of solution, ΔG_{soln} , of the two compounds:

	$\Delta H_{\text{soln}} / \text{kJ mol}^{-1}$	$\Delta S_{\text{soln}} / \text{J mol}^{-1} \text{K}^{-1}$
MgSO ₄	-91.2	-210
BaSO ₄	+26.3	-103

Use your answers to comment on the relative solubilities of the two compounds.

- (ii) Suggest a reason for the sign of ΔS_{soln} given in the table.

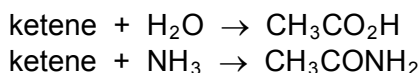
[3]

- (d) A reactive compound called ketene was discovered by N. T. M. Wilsmore at University College, London, in 1907. He passed propanone vapour over a heated platinum filament, and obtained ketene and methane as gaseous products.

- (i) A sample of 0.080 g of ketene measured at 32 kPa and 30°C occupies 150 cm³. Calculate the relative molecular mass and suggest a structural formula for ketene.

- (ii) State the shape of the ketene molecule.

- (iii) Ketene reacts rapidly with water and ammonia.



Predict the product of the reaction of ketene with ethanol.

[4]

- (e) Ketene can be converted into compound **A**, C₄H₈O₃, which has both medical and industrial applications. When **A** reacts with aqueous alkaline iodine, a yellow precipitate is formed. **A** also reacts with PCl₅ to give **B**, which rapidly reacts with water to give **C**, C₄H₇ClO₂. Treatment of **C** with alcoholic KCN, followed by hydrolysis with dilute hydrochloric acid, gives **D**. Under suitable conditions, **C** reacts with NH₃ to give compound **E**, C₄H₉NO₂. **E** is a solid with a high melting point which dissolves in water to give a nearly neutral solution. Deduce the structures for each of the compounds **A** to **E**, and explain the

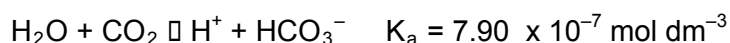
chemistry of the reactions described. [There is no need to comment on the chemistry of the formation of **A** from ketene.]

[7]
Total: [20]

- 4 (a) State **two** differences in physical properties between iron and aluminium, suggesting reasons for these differences. [3]
- (b) Explain why the atomic radius of iron is smaller than that of aluminium but the ionic radius of Fe^{3+} is bigger than that of Al^{3+} . [2]
- (c) Iron reacts with liquid bromine to form anhydrous iron(III) bromide. Anhydrous iron(III) bromide behaves in a similar manner to aluminium bromide in the reaction between benzene and bromine.
- (i) Describe the mechanism of the reaction between benzene and liquid bromine in the presence of iron.
- (ii) Like benzene, phenol also reacts with bromine, but the conditions and the products of the two reactions are different.
- Describe and account for the difference in the conditions used and the products formed in the case of phenol. [7]
- (d) Some reactions of iron are given below:
- Iron metal reacts with dilute sulfuric acid to give a pale green solution **A**, which on reaction with reagent **X** is converted to a pale yellow solution **B**. When **A** is boiled with an excess of sodium cyanide, a yellow solution **C** is produced, from which pale yellow crystals can be obtained by evaporation. A blood-red solution is obtained when a few drops of reagent **Y** is added to the solution **B**.
- (i) What are the formulae of the cations in **A** and **B**?
- (ii) Suggest identities for reagents **X** and **Y**.
- (iii) Draw the structure of the iron-containing species in **C**, showing clearly the disposition of the atoms. [5]
- (e) Finely divided iron is used as a catalyst in the manufacture of ammonia in the Haber Process.
- (i) What property of iron enables it to function as a catalyst in the reaction between nitrogen and hydrogen to form ammonia?
- (ii) Sketch, on the same axes, a clearly labeled reaction pathway diagram for the catalysed and uncatalysed Haber Process. [3]

Total: [20]

- 5 (a) The main buffering agent in blood plasma is the carbon dioxide/hydrogen carbonate system:



- (i) What is meant by the term *buffer*?
- (ii) Explain how the system above acts as a buffer when OH^- ions are added to the blood plasma.
- (iii) The pH of normal blood plasma is 7.40. Use the data above to calculate the ratio $[\text{HCO}_3^-]/[\text{CO}_2]$ and hence deduce whether blood has a greater capacity for absorbing H^+ or OH^- ions.
- (iv) During hyperventilation that comes about from severe anxiety or oxygen deficiency at high altitude the blood pH can rise above 7.40. One way to treat this problem is to breathe into a paper bag. Suggest how this works.

[6]

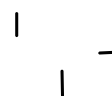
- (b) Oxygen is transported around the body in blood by haemoglobin, which is a globular protein with a quaternary structure. Chemical analysis shows that haemoglobin contains 0.34% of iron by mass.

- (i) Explain what is meant by *quaternary* structure, and how it is held together in place.
- (ii) What is the minimum possible relative molecular mass of haemoglobin?
The actual relative molecular mass of haemoglobin is four times this minimum value. What conclusion can you draw from these data?
- (iii) When glutamic acid, one of the amino acid residues in haemoglobin, is replaced by valine, the protein molecules aggregate to form insoluble polymers, causing the disease known as sickle cell anaemia. The structures of glutamic acid and valine are given below. Suggest the type of intermolecular interactions which causes the aggregation of the protein molecules.

Glutamic acid



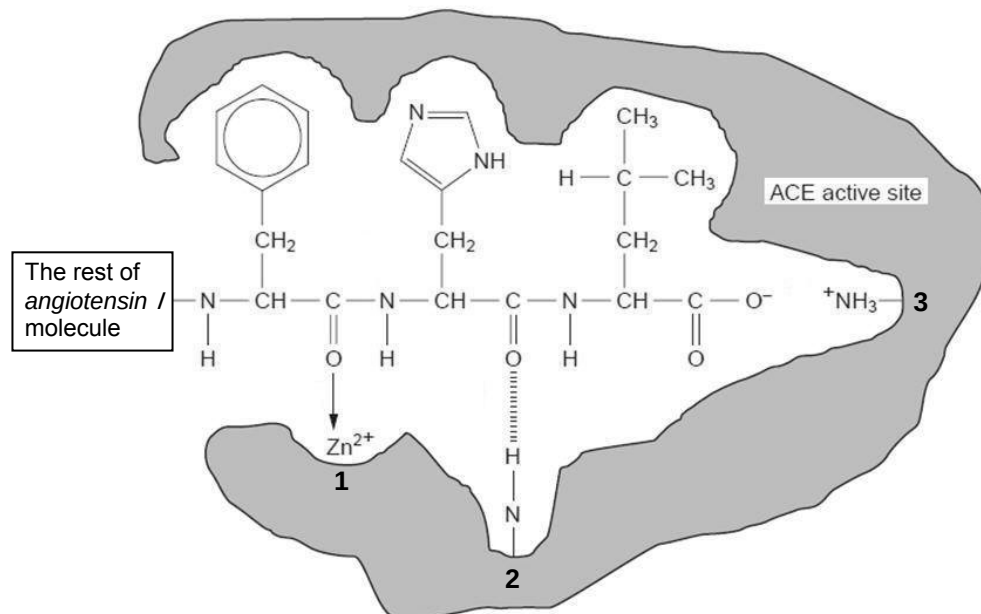
Valine



[4]

- (c) It has been discovered that high blood pressure is caused by an overproduction of *angiotensin II*, a dipeptide, in the body. *Angiotensin II* is made from a polypeptide *angiotensin I* by the action of an enzyme known as ACE.

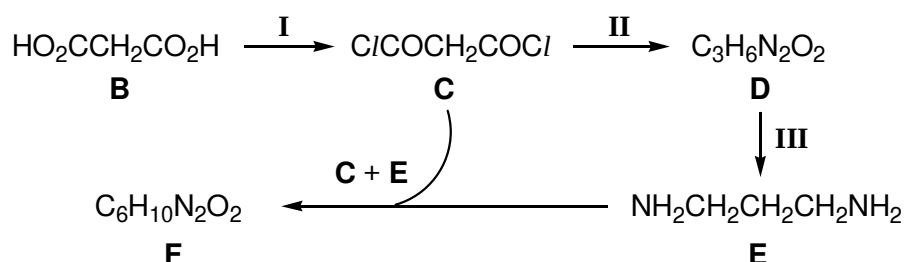
The diagram shows one end of an *angiotensin I* molecule bound to the active site of ACE.



- Explain how *angiotensin I* is bonded to the three numbered groups, 1, 2 and 3, shown on the active site of the enzyme ACE.
- State the type of reaction which occurs when *angiotensin II* is made from *angiotensin I*, and draw the structure of *angiotensin II*.
- ACE inhibitors are small peptides used as medicine to prevent the overproduction of *angiotensin II*. Suggest why small peptides are not effective medicines when taken orally.

[5]

- (d) Propanedioic acid, **B**, is the starting compound in the following reaction scheme.



- Suggest reagents for Steps I and II.
- Name the functional group present in **D**.
- What type of reaction occurs in Step III?
- Draw the displayed formula of **F** which is formed between one mole of **C** and one mole of **E**.

[5]

Total: [20]

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